Price Searchers

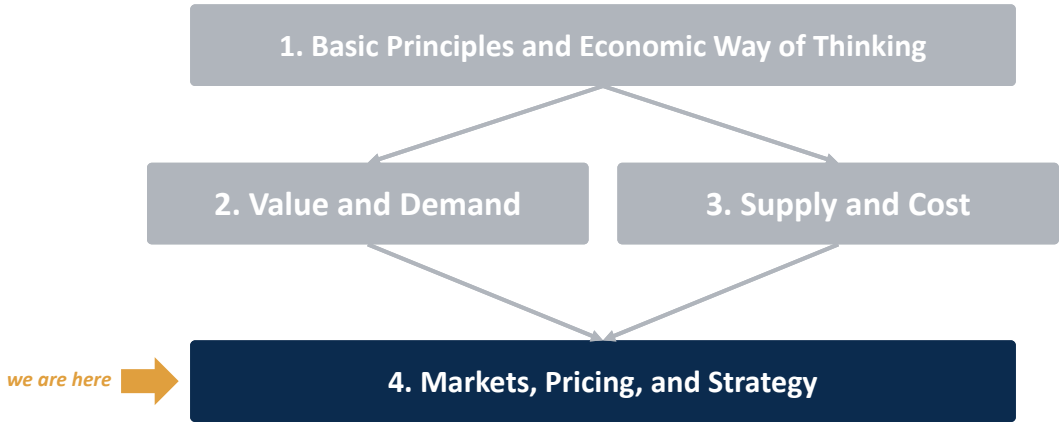
Monopoly and Monopolistic Competition

Management 405

Prof. Nico Voigtländer · UCLA Anderson

1

Agenda for the Class



2

Module 5 · Agenda

Outline of Module 5

1. **Profit Maximization for Price Searchers** — In class
   How a firm with market power picks its price and quantity
   - 1b **Markup and Lerner Index** — In class
     How far price sits above marginal cost
2. **Monopoly and Welfare Impact** — In class
   Monopoly profits come at consumers' expense
3. **Rising Market Power and Regulations** — In class
   Rising market concentration and government responses
4. **Monopolistic Competition** — In class
   Many firms, differentiated products: entry erodes profits

3

Module 5 · Market Structures

The Four Basic Market Structures

Use the same analytics to find optimal price/quantity

	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of firms	Many	Many	Few	One
Type of products sold	Identical	Differentiated	Identical or Differentiated	"Unique"
Barriers to entry	None	None	Some	Many

More firms, more similar products, easier entry, more competition

4

Module 5 · Agenda

Outline of Module 5

1 **Profit Maximization for Price Searchers** — In class
How a firm with market power picks its price and quantity

1b Markup and Lerner Index — In class

2 Monopoly and Welfare Impact — In class

3 Rising Market Power and Regulations — In class

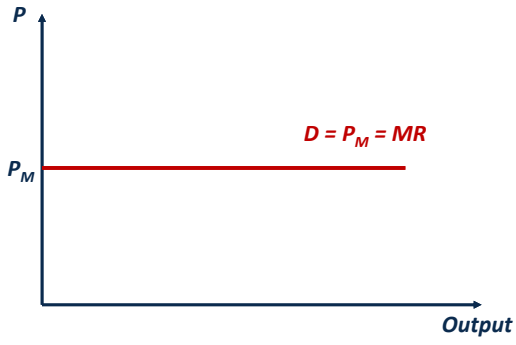
4 Monopolistic Competition — In class

5

Module 5 · Profit Maximization for Price Searchers

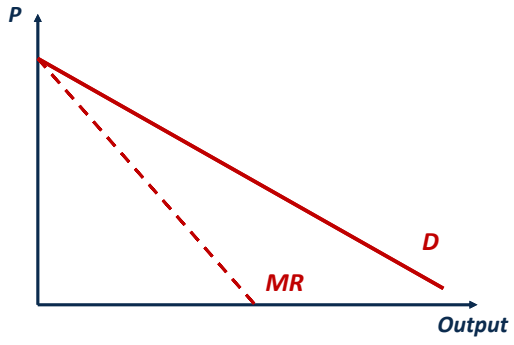
Firm Demand: Price Takers vs. Price Searchers

Price taker



The demand curve faced by a price taker is the market price (which in turn equals *MR* for a price taker)

Price searcher



The demand curve faced by a price searcher is downward-sloping
Marginal revenue (MR) is different from the demand curve (D)

6

Test of Market Power

- If we raise our price, will we lose all our customers?
  - "yes" → Price **taker**
    - We have no market power
  - "no" → Price **searcher**
    - We have market power!
    - We face a downward-sloping demand curve

7

Profit Maximization for Price Searcher

- Profit maximization for price searchers:

$$\text{MR} = \text{MC}$$

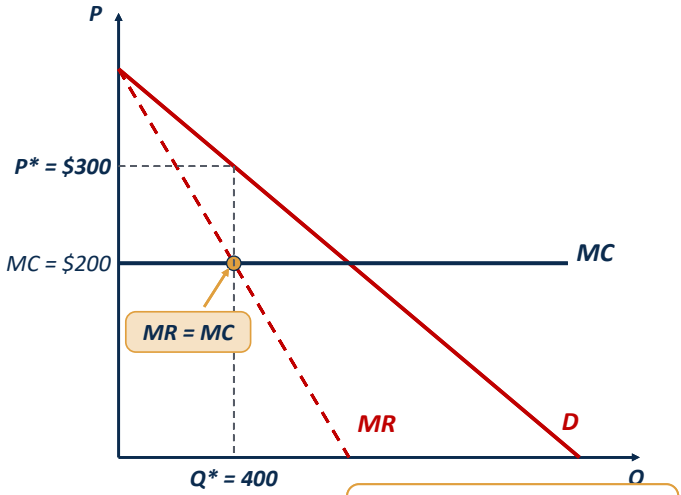
$$Q = 1{,}600 - 4P$$

(estimated demand curve)

$$\rightarrow P = 400 - 0.25Q$$

(inverse demand → what we plot)

MR is derived from the (inverse) demand curve



Teaching Note: MR = MC

8

MR = MC: Computation Example

1. **Demand and inverse demand**

$$Q = 1{,}600 - 4P \quad \Rightarrow \quad P = 400 - 0.25Q$$

2. **Total and marginal revenue**

$$\text{TR} = P \cdot Q = 400Q - 0.25Q^2 \quad \Rightarrow \quad \text{MR} = 400 - 0.5Q$$

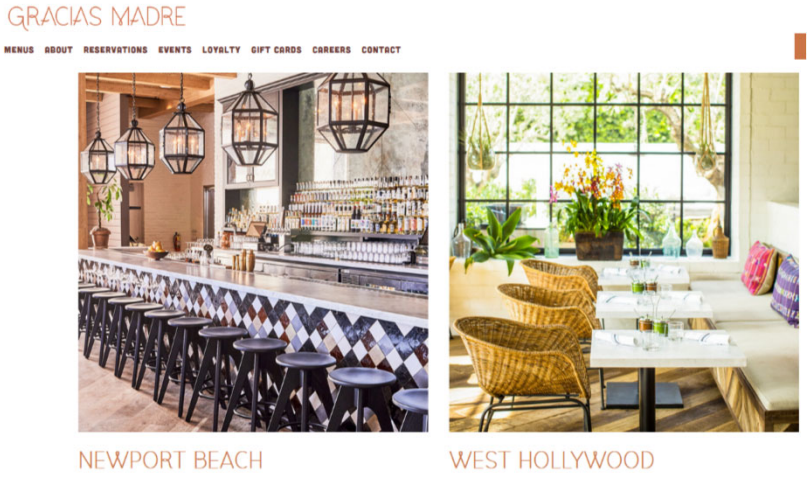
3. **Set MR = MC (MC = 200)**

$$400 - 0.5Q = 200 \quad \Rightarrow \quad Q^* = 400$$

4. **Optimal price: go up to the demand curve**

$$P^* = 400 - 0.25 \cdot 400 = \$300$$

Profit Maximization of a Restaurant



Practice Video: Optimization of a Price Searcher

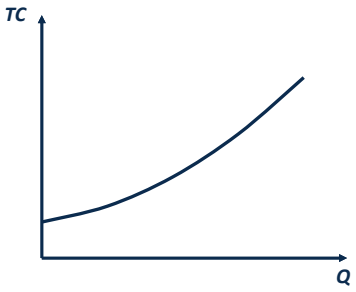
This Is from the Practice Video: Profit Maximization for a Restaurant

GRACIAS MADRE

- The restaurant demand (on a daily basis) is:
$$P = 60 - 0.1Q$$
- The total cost is (also daily; fixed cost = rent of $2,000):
$$\text{TC} = 2{,}000 + 10Q + 0.15Q^2$$

- **What are *P** and *Q**? What are profits?**

In class, we will focus on the graphics/intuition. **Watch the Practice Video for exam prep!**



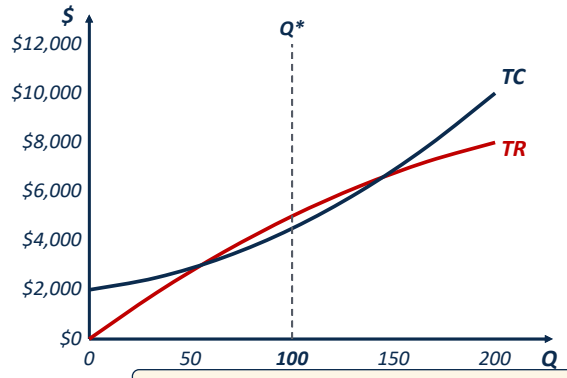
Practice Video: Optimization of a Price Searcher

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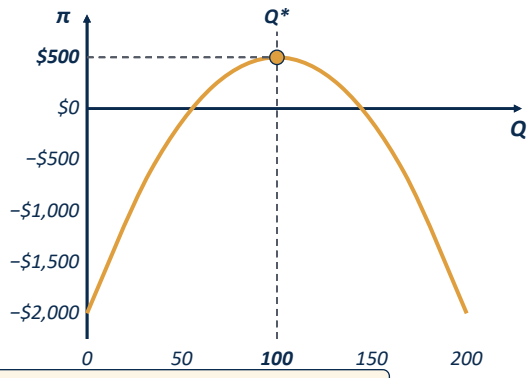
Illustration of the Practice Video Example: Profit Maximization

GRACIAS MADRE

***Total* Revenue and *Total* Cost**



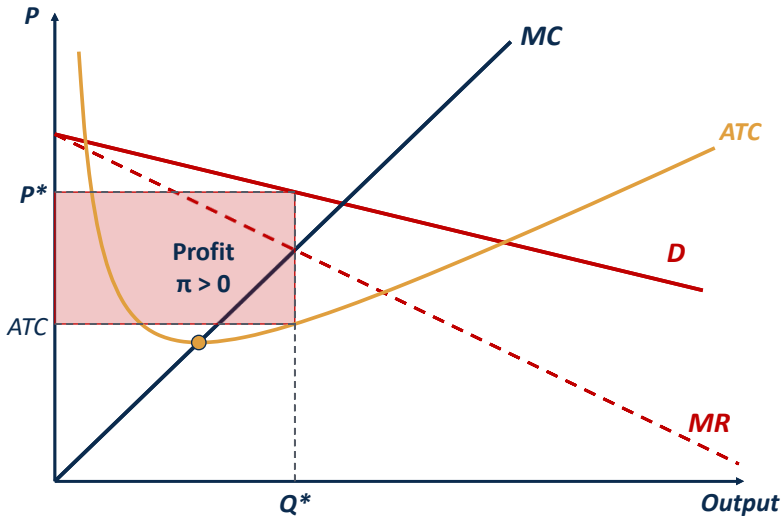
Profit



To find the profit-maximizing quantity, use marginal analysis (***MR = MC***)

12

Price Searcher: Maximizing Profit Using MR=MC



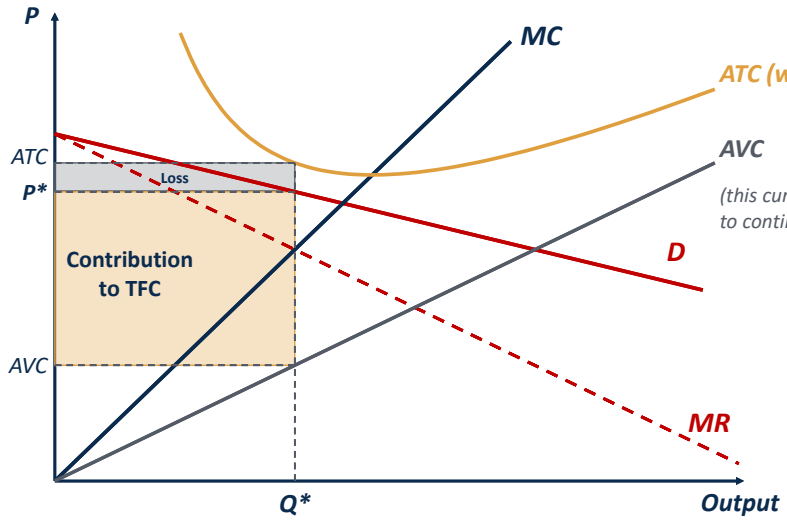
GRACIAS MADRE



Now suppose: rent increases → *ATC* moves up (next slide)

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Price Searcher: Minimizing Loss



GRACIAS MADRE

ATC (with higher rent)

AVC

(this curve is needed to check whether to continue producing in the short run)

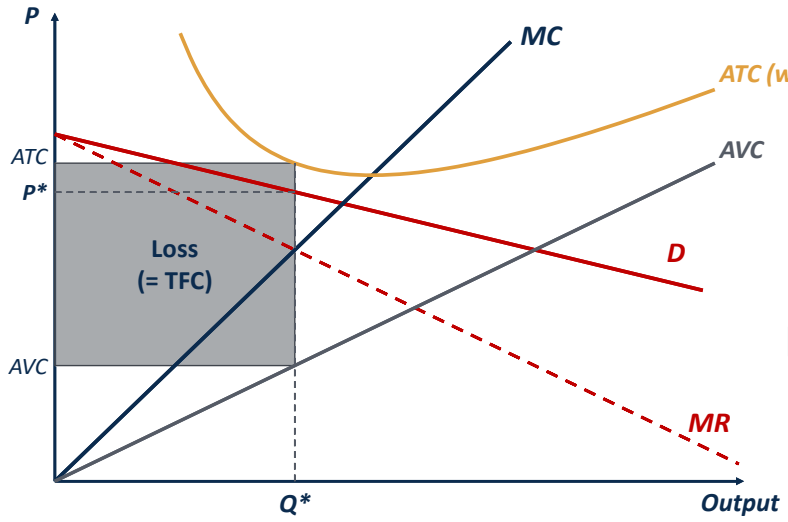
Losses, because P* < ATC

But P* > AVC → the restaurant should continue to produce, to cover part of its TFC

TFC = (ATC − AVC) · Q

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Loss When Stopping to Produce in the Short Run

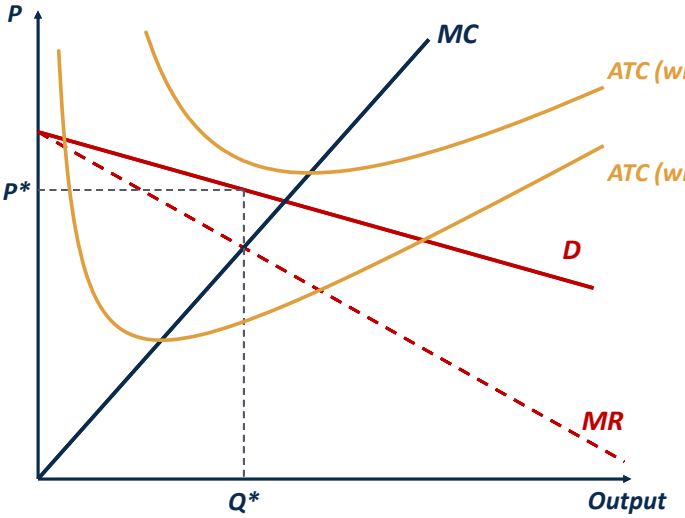


GRACIAS MADRE

If the restaurant stops serving customers but still has to pay the rent, it loses the whole fixed cost (TFC), which is equal to the rent

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Price Searcher: Fixed Cost Does Not Impact Price



GRACIAS MADRE

P does not change if fixed costs go up*

When fixed costs (and thus *ATC*) increase, *P** **does not change. But profits fall**

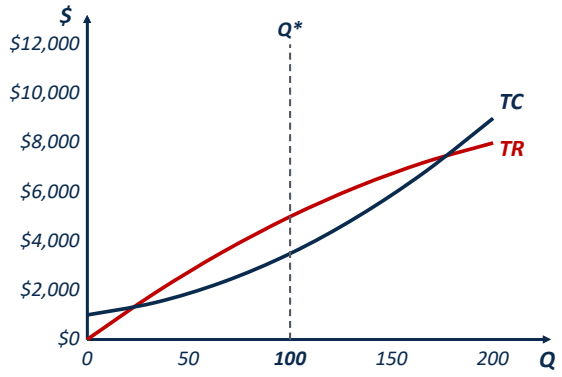
16

Illustration Based on the Practice Video Example

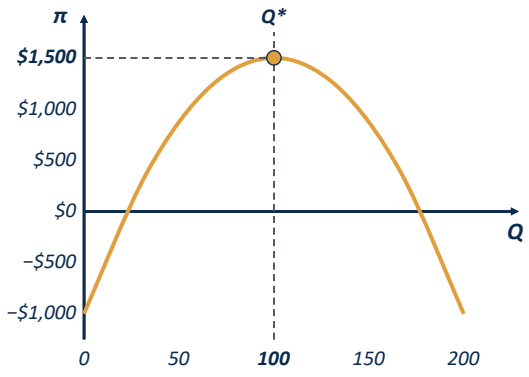
Case 1: **Low Fixed Cost** (Rent $1,000)

GRACIAS MADRE

Total revenue and total cost



Profit



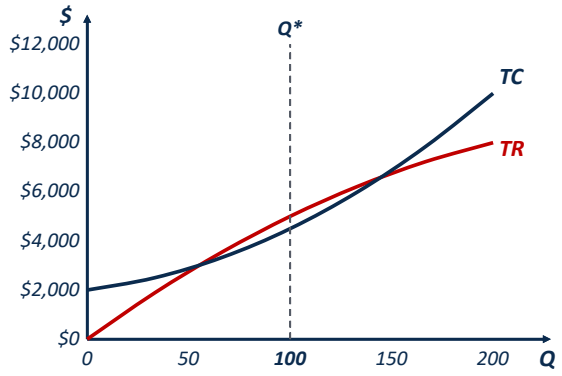
17

Illustration Based on the Practice Video Example

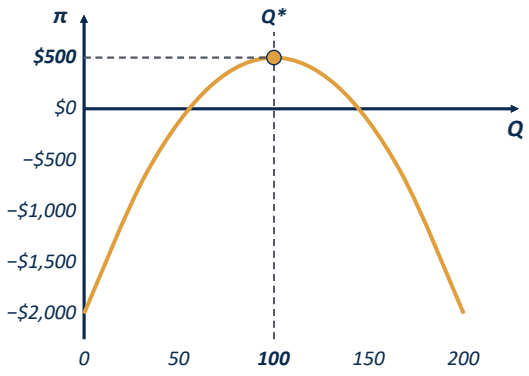
Case 2: **Medium Fixed Cost** (Rent $2,000)

GRACIAS MADRE

Total revenue and total cost



Profit



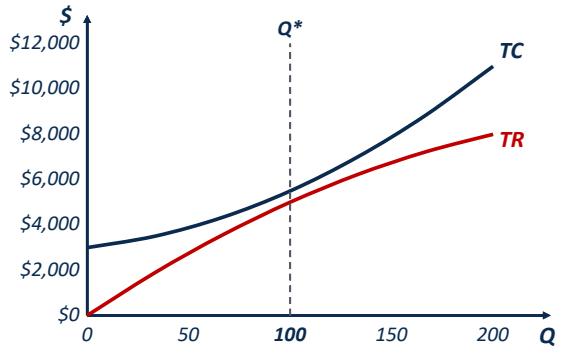
18

Illustration Based on the Practice Video Example

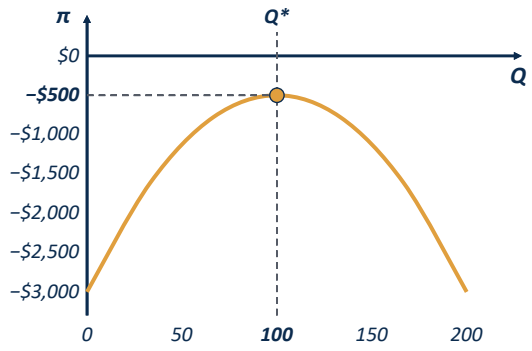
Case 3: **High Fixed Cost** (Rent $3,000)

GRACIAS MADRE

Total revenue and total cost



Profit



In all three fixed-cost cases, the profit-maximizing quantity remains the same

Additional Exercise in the Practice Video

GRACIAS MADRE

Should Gracias Madre buy a liquor license?

Practice Video: Optimization of a Price Searcher



Outline of Module 5

1. Profit Maximization for Price Searchers — In class

1b Markup and Lerner Index — In class
How far price sits above marginal cost

2. Monopoly and Welfare Impact — In class
3. Rising Market Power and Regulations — In class
4. Monopolistic Competition — In class

Markup and the Lerner Index

- A firm charges a markup if its price is above its marginal cost ($P > \text{MC}$)
  - A common definition is Markup = $\frac{P}{\text{MC}}$
- We will follow the textbook and use the **Lerner Index**: a measure of the market power of a firm:

$$\text{Lerner Index} = \frac{P - \text{MC}}{P}$$

  - The percentage of a firm's price that is due to a mark-up above its marginal cost
    - E.g., P = 2; MC = 1.5 → 25% of the price is due to mark-up
  - The Lerner Index is the larger the higher is the markup
  - Between 0 and 1
    - Zero under perfect competition ($P = \text{MC}$)
    - Increases with market power (e.g., advertising, M&A)

Lerner Index for Major US Companies

Percentage of Price that is due to Markup

$$\text{Lerner Index} = \frac{P - \text{MC}}{P}$$

Company	Lerner Index (P – MC) / P
Alphabet	0.55
Amazon.com	0.33
American Airlines	0.24
Apple	0.33
AT&T	0.35
Johnson & Johnson	0.72
Microsoft	0.56
Pfizer	0.78
Tesla	0.25
United Airlines	0.21
Whole Foods Market	0.19

Source: De Loecker et al. (2019)

Lerner Index

- The Lerner index is related to the firm's elasticity:

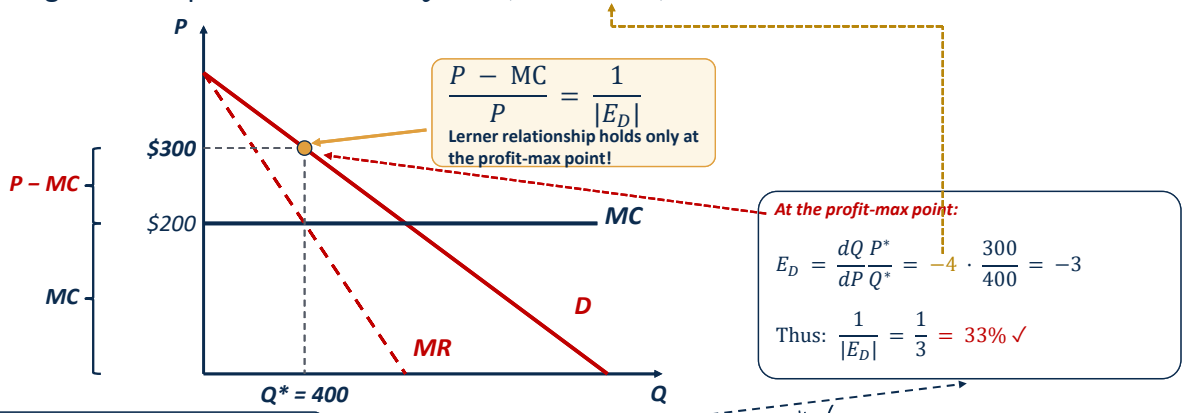
The elasticity of demand has a negative value

$$\frac{P - \text{MC}}{P} = -\frac{1}{E_D} = \frac{1}{|E_D|}$$

- As demand becomes more elastic, the % of markup in P falls
  - Note: if $|E_D| \rightarrow \infty$ then Lerner Index = 0
- Important: This relationship **holds only at the profit-maximizing point** (P^*, Q^*)

Illustrating the Lerner Index

Using the example from above: $Q = 1{,}600 - 4P$; $\text{MC} = 200$



$$\frac{P - \text{MC}}{P} = \frac{1}{|E_D|}$$

Lerner relationship holds only at the profit-max point!

At the profit-max point:

$$E_D = \frac{dQ}{dP}\frac{P^*}{Q^*} = -4 \cdot \frac{300}{400} = -3$$

Thus: $\frac{1}{|E_D|} = \frac{1}{3} = 33\%$ ✓

$$\frac{P - \text{MC}}{P} = \frac{300 - 200}{300} = 33\%$$

→ 33% of the price is due to markup

Check: The two formulas give the same result ✓

Lerner Index – Relationship to Accounting *(For Your Notes Later, When You Take Accounting)*

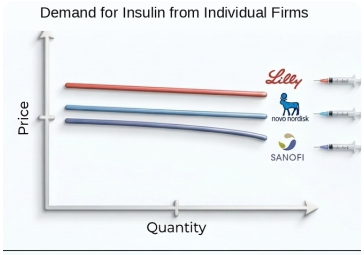
- In accounting, you'll see:
  - Gross Margin = $\frac{\text{TR} - \text{COGS}}{\text{TR}} \approx \frac{\text{TR} - \text{TVC}}{\text{TR}}$ ("Cost of Goods Sold" ≈ *TVC*)
  - Divide by quantity: $\frac{\text{TR} - \text{TVC}}{\text{TR}} = \frac{P \cdot Q - \text{AVC} \cdot Q}{P \cdot Q} = \frac{P - \text{AVC}}{P}$ (*Average variable cost*)

→ **Gross Margin** ≈ $\frac{P - \text{AVC}}{P}$ Lerner Index: $\frac{P - \text{MC}}{P}$

- Thus: The Gross Margin is similar to the Lerner Index, but it is measured at <u>average</u> instead of <u>marginal</u> cost
  → Economists think at the <u>margin</u>

Firm and Market Elasticity

- Note that the Lerner index is calculated based on the firm elasticity
- Important:
  - Firm and market elasticity are generally different
  - **Firm elasticity tends to be larger than market elasticity,** especially if there are many firms competing in an industry
  - Across firms, those with more differentiated products tend to face lower elasticities





Mark-Up and Elasticity

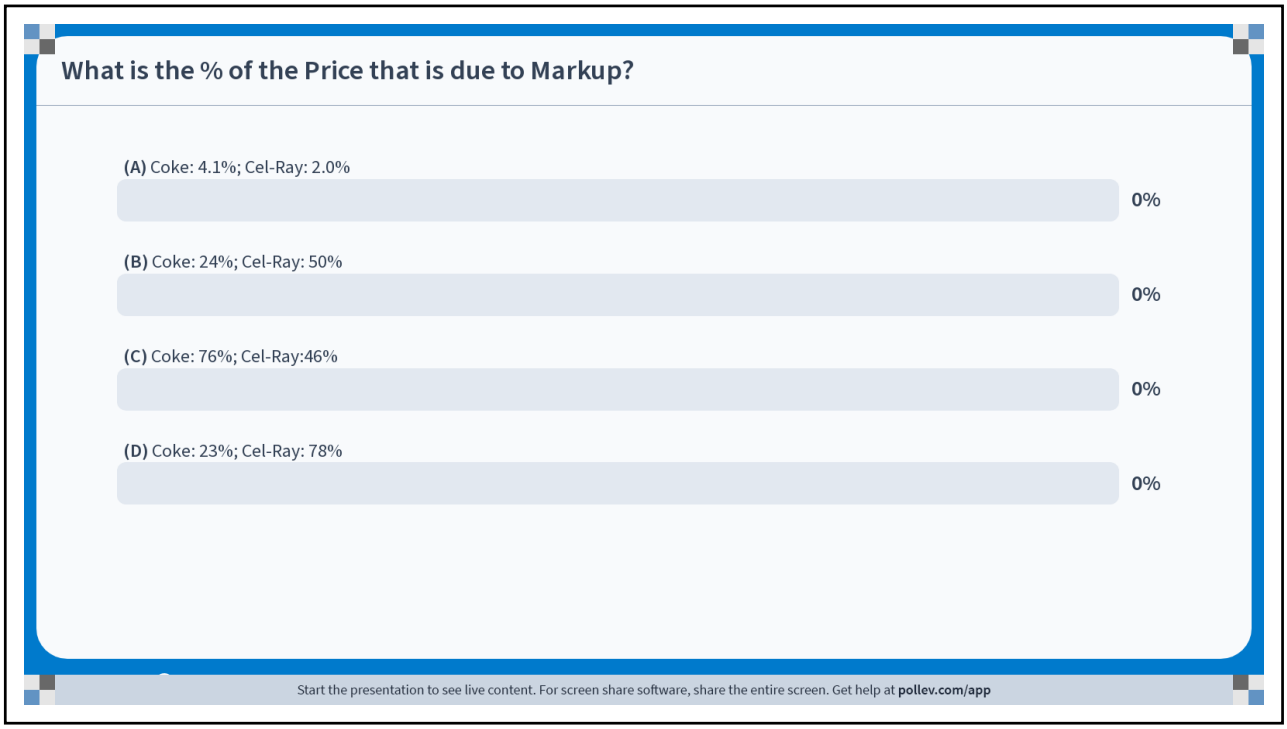
- Coca-Cola's demand elasticity is −4.1
- Dr. Brown's Cel-Ray has a demand elasticity of −2.0
- What is the % of the price that is due to markup for the two products?
  - Note: You may assume that elasticities are for the profit-maximizing P and Q





Robert Takata/ Guzzle & Nosh

Poll Break



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Solution

Poll Break

30

Price Searcher Take-Aways

- Basic rule for profit maximization:
  - Set *MR = MC* to find *Q**
  - Continue to produce in the short run if *P ≥ AVC*
- Key difference to price taker:
  - The price searcher sets *P* > MC*
- Mark-up depends on elasticity

Outline of Module 5

1	Profit Maximization for Price Searchers	In class
1b	Markup and Lerner Index	In class
2	**Monopoly and Welfare Impact** Monopoly profits come at consumers' expense	In class
3	Rising Market Power and Regulations	In class
4	Monopolistic Competition	In class

Where We Are and Where We Are Going

Least Market Power ⟷ **Most Market Power**

Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
(Price Takers)	(Price Searchers)		
		BOEING / AIRBUS	LA DWP
Commodity markets	*Restaurants in WeHo*	*Wide-body aircraft*	*Municipal utilities*
			TODAY

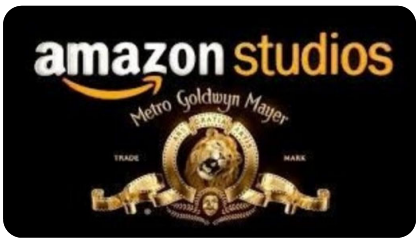
Market Structure Characteristics

MONOPOLY

	Monopoly
Number of actual or potential competitors	Single seller
Product differentiation	Very high – no close substitutes
Entry conditions	Restricted access (barriers)
Profit potential	Profit potential in short and long run
Examples	Microsoft (before the rise of Apple); gas, water and electric utilities

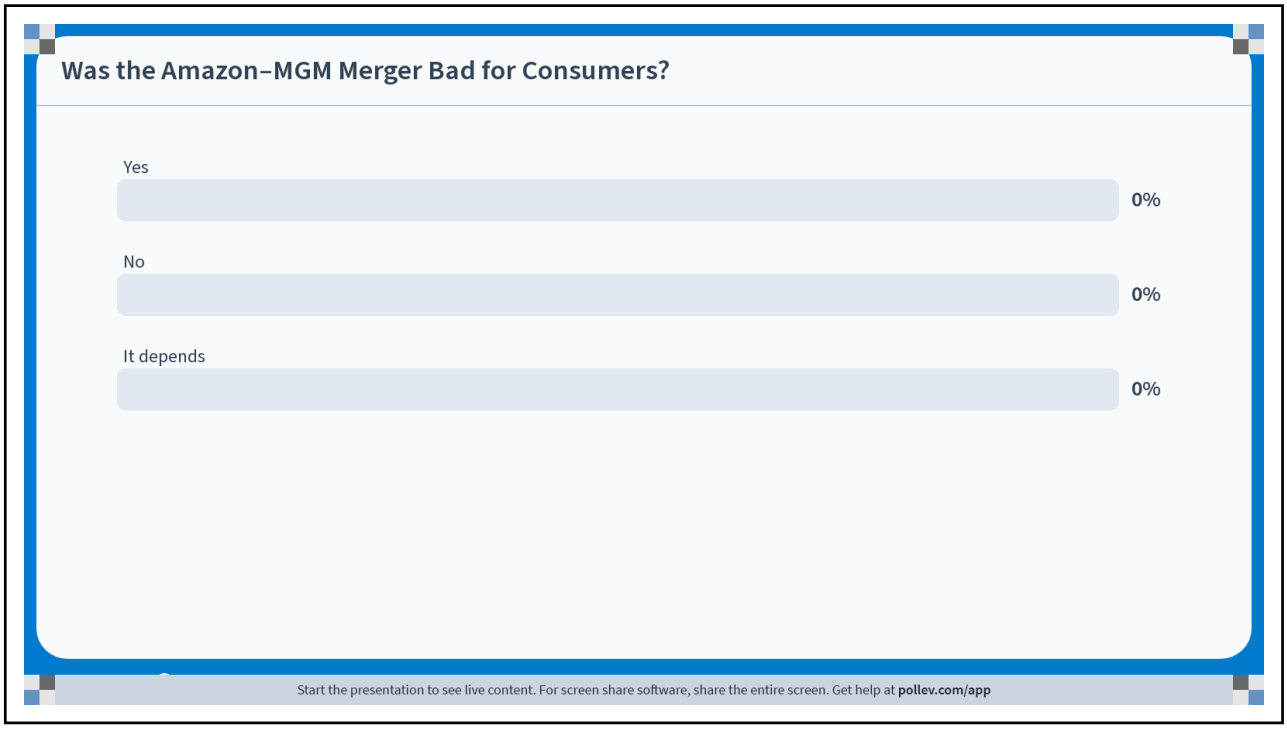
Poll: Was the Amazon–MGM Merger Bad for Consumers?

- In March 2022, Amazon completed its acquisition of MGM Studios for $8.45 billion
- Was this merger bad for consumers, or did it not harm them?



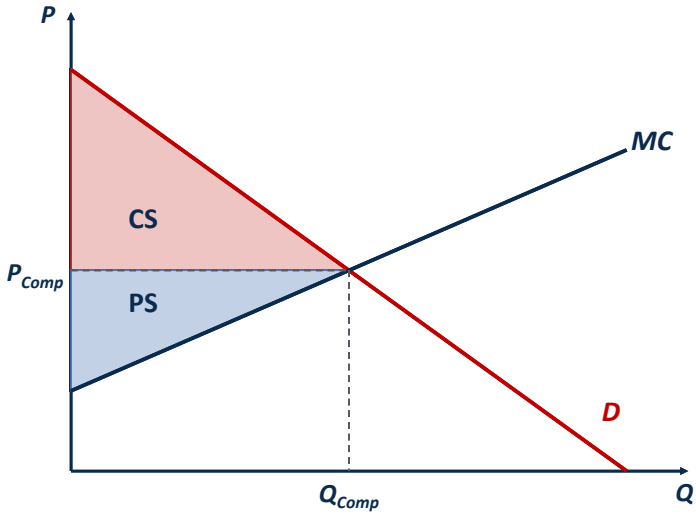
Poll Break

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Recall: Consumer and Producer Surplus Under Competition

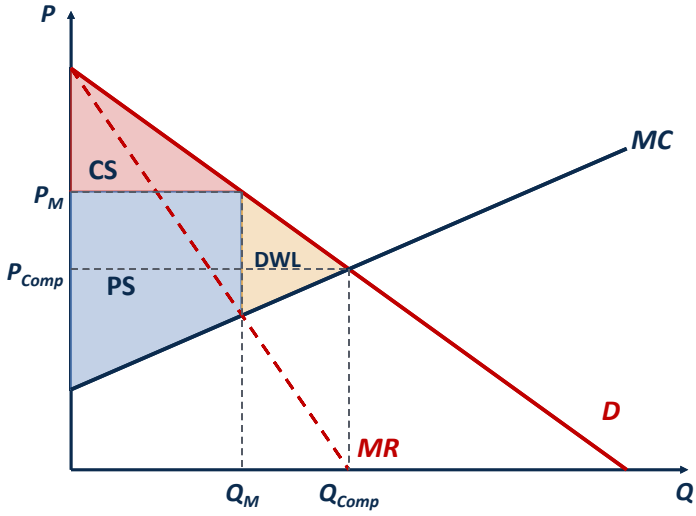


Consumer surplus (CS) under competition

Producer surplus (PS) under competition

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Monopoly: Good for Business, Bad for Consumers and Welfare



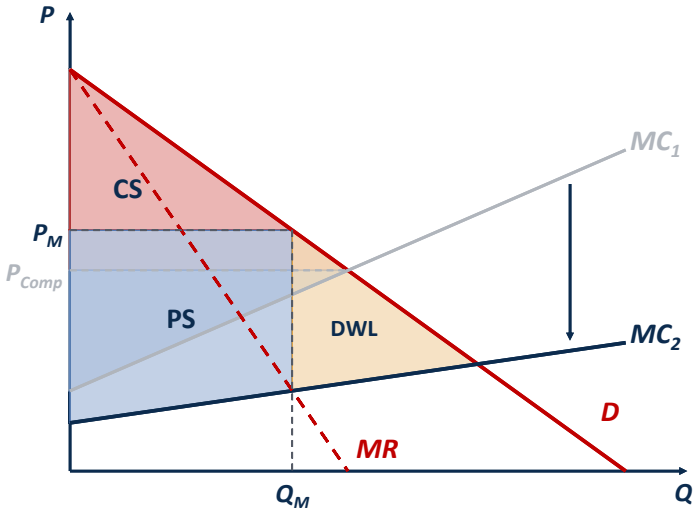
Consumer surplus (CS) under monopoly

Producer surplus (PS) under monopoly

Deadweight loss (DWL) under monopoly

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What If Costs Are Lower for One Large Producer?



CS and PS under competition (original MC)

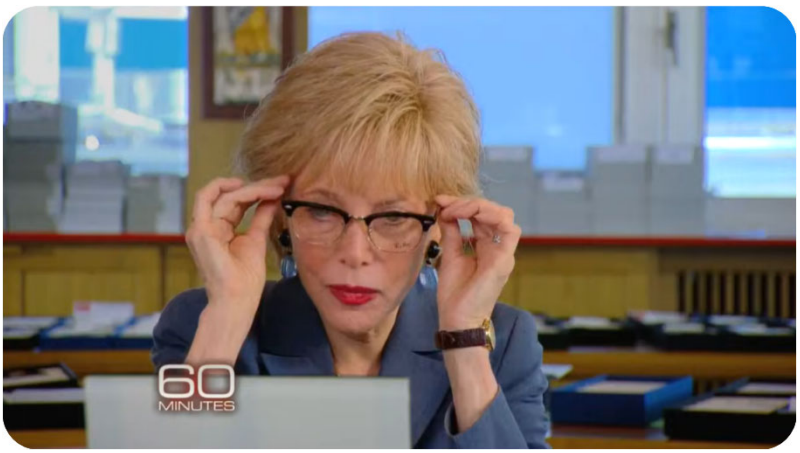
Consumer surplus (CS) under monopoly with lower MC

Producer surplus (PS) under monopoly with lower MC

Deadweight loss (DWL) under monopoly with lower MC

Lower costs after mergers can benefit consumers, but only if higher market power does not reverse these gains

The Case of Luxottica (Now EssilorLuxottica)



60 Minutes on Luxottica – watch on YouTube (from 9:49)

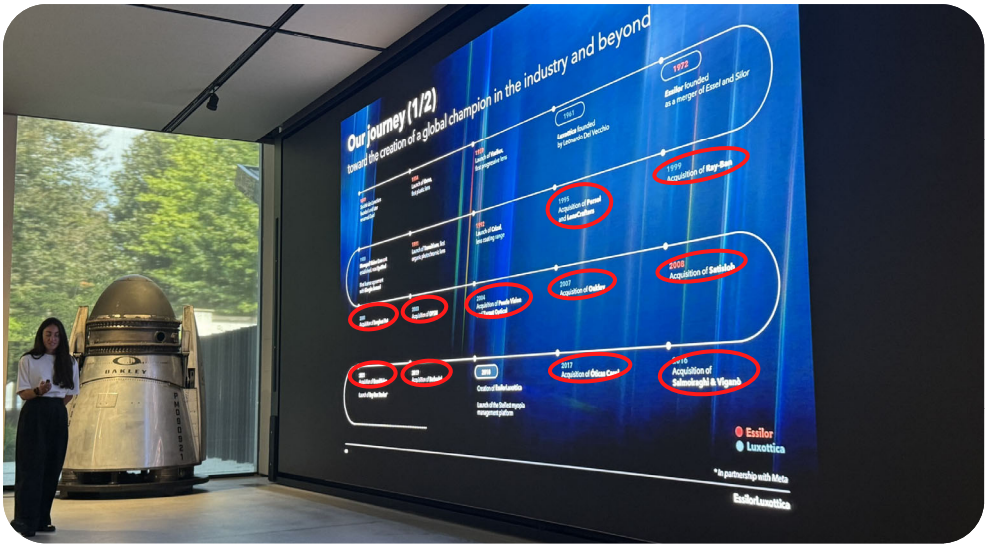
Class Discussion

So I Visited EssilorLuxottica with an EMBA IBR Class…



41

…and This Was Their Chart on Company History



42

The Case of Luxottica

EssilorLuxottica

- Which type of market is Luxottica operating in?
- How do they achieve market power?
- Do they have a high markup?
- How does the following statement by Luxottica's former CEO relate to what we learned about the demand curve?
  - *« Everything is worth what people are ready to pay »*

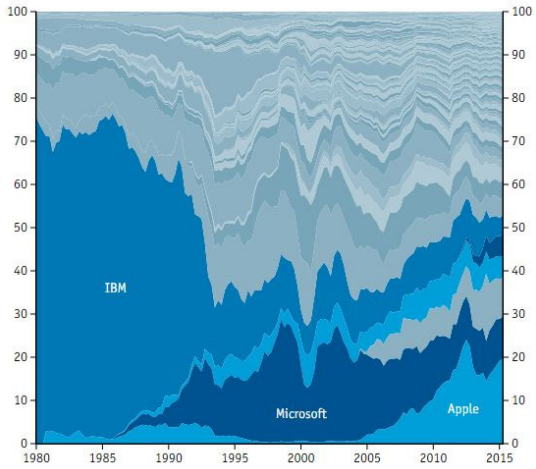


Monopoly Take-Aways

- Monopolist: Large degree of market power
  - Enables high profits
  - Reduces consumer surplus
  - Reduces overall (society) welfare
- Silver lining: Patents, licenses and copyrights
  - The prospect of monopoly power provides incentives for innovation (e.g., new drugs)
  - Consumer surplus from new product creation can outweigh deadweight loss due to high prices



All Firms Face Potential (Future) Competitors



Source: The Economist, economist.com/techfirms

Ultra-Low-Cost Carrier Airlines: Entry Drives Down Prices

- Ultra-low-cost carriers (ULCC)
- On routes that ULCC start to fly:
  - Fares about 21% lower than average[1]
  - Number of passengers goes up
- Even the threat of entry lowers fares:
  - Once Southwest served both endpoints of a route, incumbents cut fares; over half of the drop came before Southwest entered[2]



[1] Bachwich, A. R., and M. D. Wittman (2017): "The Emergence and Effects of the Ultra-Low Cost Carrier (ULCC) Business Model in the U.S. Airline Industry," Journal of Air Transport Management 62: 155–164.
[2] Goolsbee, A., and C. Syverson (2008): "How Do Incumbents Respond to the Threat of Entry? Evidence from the Major Airlines," Quarterly Journal of Economics 123(4): 1611–1633.

Module 5 · Agenda

Outline of Module 5

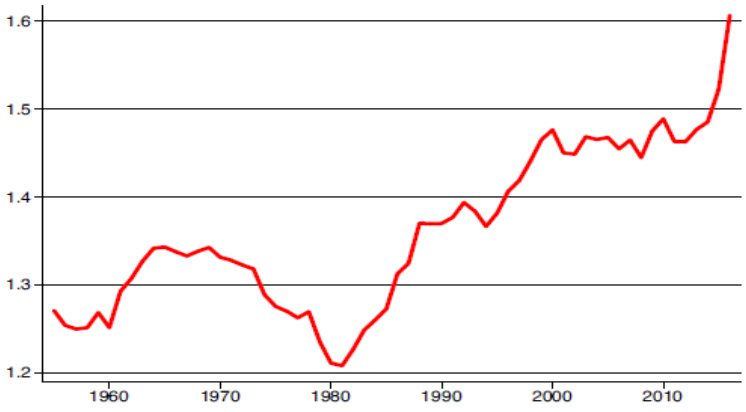
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   - 1b. Markup and Lerner Index — In class
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Module 5 · Rising Market Power and Regulations

Rising Markups in the US

Average markup (P/MC) in the US economy*

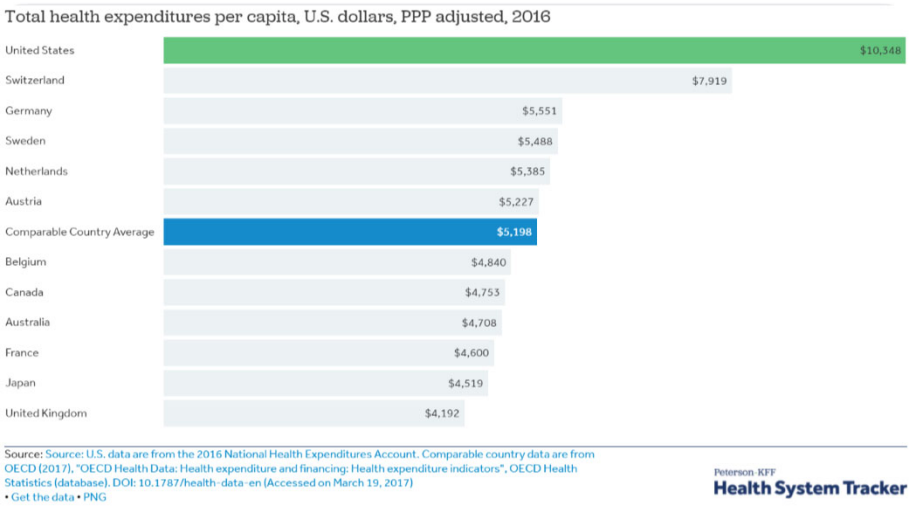


Source: De Loecker et al., QJE 2020: The Rise of Market Power and the Macroeconomic Implications
**The source uses the markup definition (P/MC) – this is different from the Lerner index (P – MC)/P*

Note: If markup = 1.5 → Lerner index = 0.33

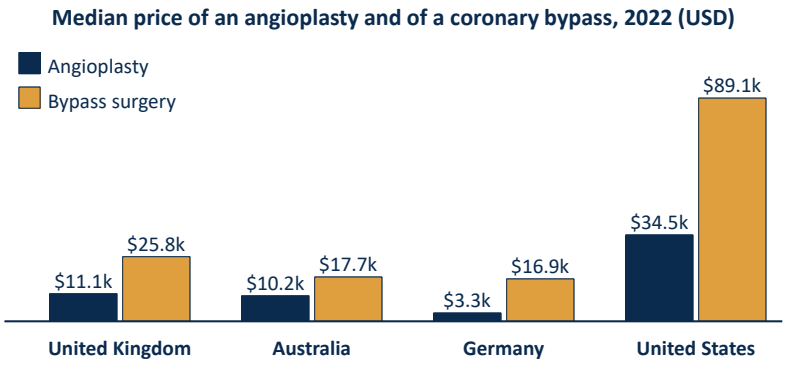
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Health Care Expenditures Across Countries



Health Care Cost Comparison: Heart-Related Treatments

- Angioplasty: a procedure used to open blocked coronary arteries
- Bypass surgery: redirect blood around a blocked section of an artery in the heart



Source: iFHP and HCCI, International Healthcare Cost Comparison Report 2024 (2022 data); median private-insurer claims costs

Concentration in US Industries



A market is "concentrated" if CR4 (the share of sales of the top-4 companies) is above 50%

concentrationcrisis.openmarketsinstitute.org

Government Regulations to Control Market Power

- Price regulations
  - Happens frequently with natural monopolies: electricity, natural gas, (historically: local telephone service)
- Antitrust laws
  - Scrutiny on mergers/acquisitions that would limit competition
  - FTC successfully blocked Kroger's acquisition of Albertsons in 2024 (~ $24.6 billion)
  - Ban collusion on prices
- Nevertheless: Very high and rising market concentration in the US



Killer acquisitions of startups

Module 5 · Agenda

Outline of Module 5

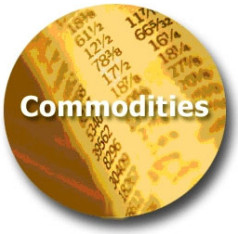
1. Profit Maximization for Price Searchers — In class
   - 1b. Markup and Lerner Index — In class
2. Monopoly and Welfare Impact — In class
3. Rising Market Power and Regulations — In class
4. **Monopolistic Competition** — In class
   Many firms, differentiated products: entry erodes profits

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Module 5 · Monopolistic Competition

Where We Are

Least Market Power ⟷ **Most Market Power**

Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
(Price Takers)	(Price Searchers)		
		BOEING / AIRBUS	LA DWP
Commodity markets	*Restaurants in WeHo*	*Wide-body aircraft*	*Municipal utilities*

TODAY

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Market Structure Characteristics

	Monopolistic Competition
Number of actual or potential competitors	Many sellers
Product differentiation	Perceived differences but relatively close substitutes *(How to get market power: Palessi)*
Entry conditions	No (major) barriers to entry
Profit potential	Normal economic profit in long run; losses and profits in short run only
Examples	Professional services (plumbers, gardeners); restaurants, gas stations; clothing

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Monopolistic Competition Example: Gas Stations

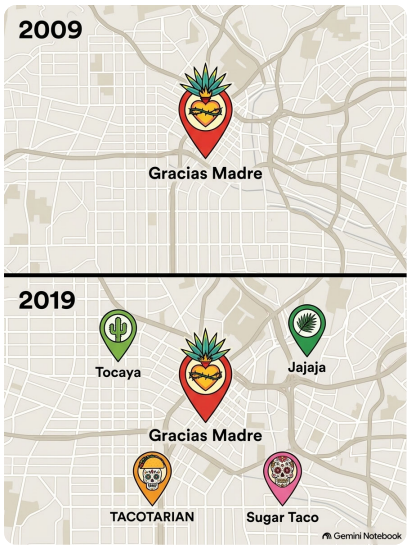


- Product differentiation by location

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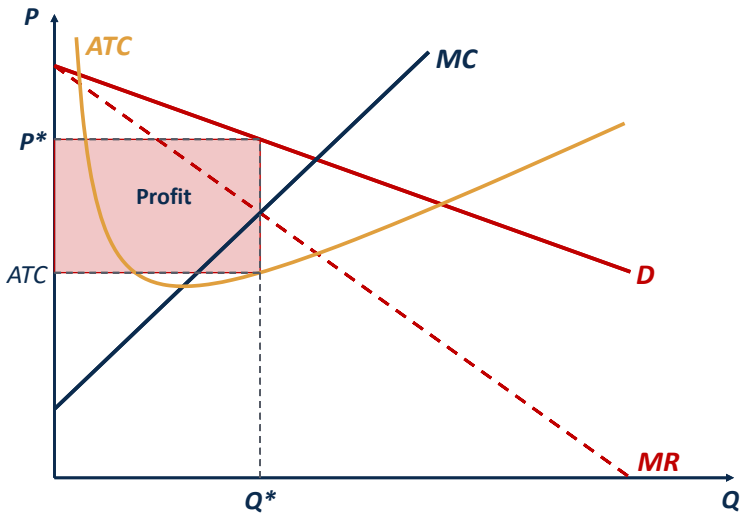
Monopolistic Competition: The Role of Entry

- Monopolistic competition
  - Large number of firms
  - No (major) entry barriers
- **Short run**: economic profits possible
- **Long run**: entry erodes economic profits



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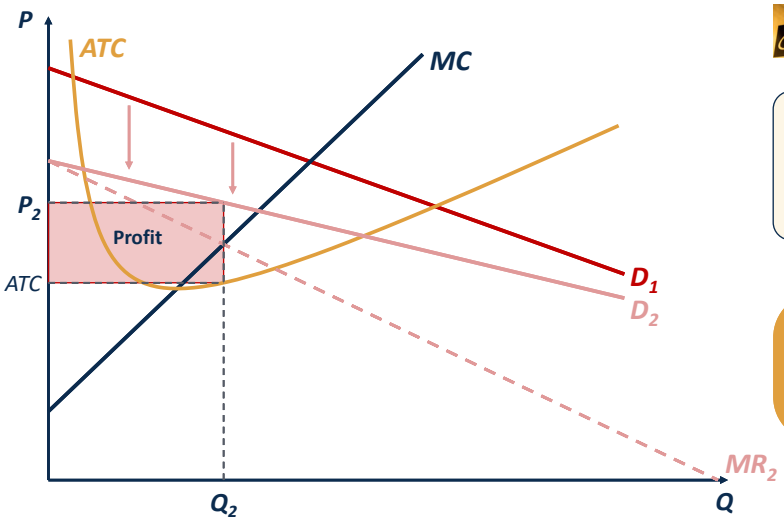
Monopolistic Competition: Short Run



GRACIAS MADRE

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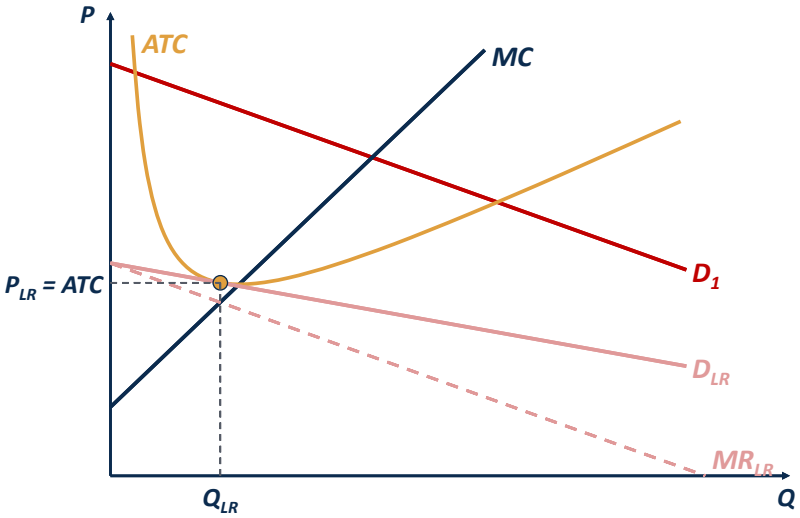
Monopolistic Competition: Entry of New Firms



New entrants → the incumbent's demand curve shifts down and becomes more elastic (D_2)

Profits of the incumbent firm decline as more competitors with similar (but differentiated) products enter

Monopolistic Competition: Long-Run Equilibrium



Entry shifts demand down and makes it more elastic (flatter)

In the long run, each monopolistically competitive firm earns zero economic profit (P = ATC)

Note: This holds under the assumption that firms are symmetric in costs but produce differentiated products

Barriers to Entry

- Entry erodes profits in a monopolistically competitive market
- Different from monopoly: A monopolist makes positive economic profits
- Why the difference?
  - **Barriers to entry** protect a monopolist
  - Can firms create barriers to entry?

"Structural" Barriers to Entry

- Legal barriers
- Network effect
- Economies of scale
- Learning
- Control of resources











Switching Costs Can Also Raise Barriers to Entry

Loyalty programs



Compatibility



Learning

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